

Quantization and Speech Coding

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EQ2321 Speech and Audio Signal Processing

Overview

- Section 1: uniform scalar quantizer
- Section 2: parametric coding of speech
 - frame energy quantization
 - pitch and voiced/unvoiced quantization
 - LP parameter VQ
 - bit-allocation optimization
- Section 3: speech waveform quantization
- Section 4: adaptive open-loop DPCM

Section 1 – Q1: Uniform scalar encoder and decoder

The implemented midrise quantizer uses

$$\Delta = \frac{2x_{\max}}{2^R}, \quad i = \left\lfloor \frac{x - (m - x_{\max})}{\Delta} \right\rfloor, \quad \hat{x} = m - x_{\max} + \left(i + \frac{1}{2}\right) \Delta.$$

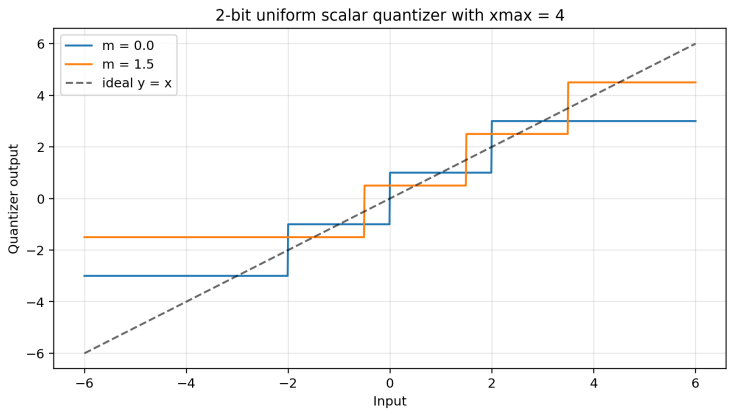
Indices are clipped to the valid interval $0, 1, \dots, 2^R - 1$.

Result

- The encoder maps each sample directly to a quantization interval.
- The decoder reconstructs the midpoint of the selected interval.
- The per-sample complexity is independent of the number of levels because the method uses direct arithmetic instead of a search over codebook entries.

Section 1 – Q2: Quantizer characteristic for $R = 2$,

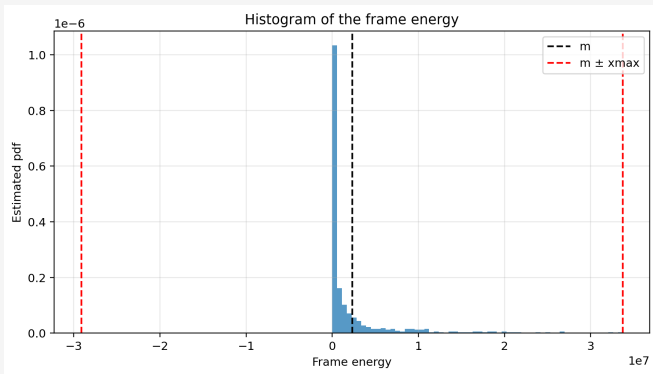
$$x_{\max} = 4$$



Interpretation

For $m = 0$, the staircase is centered at the origin. For $m = 1.5$, the entire characteristic shifts to the right, exactly as required by the offset definition.

Section 2.1 – Q1: Histogram of the frame energy

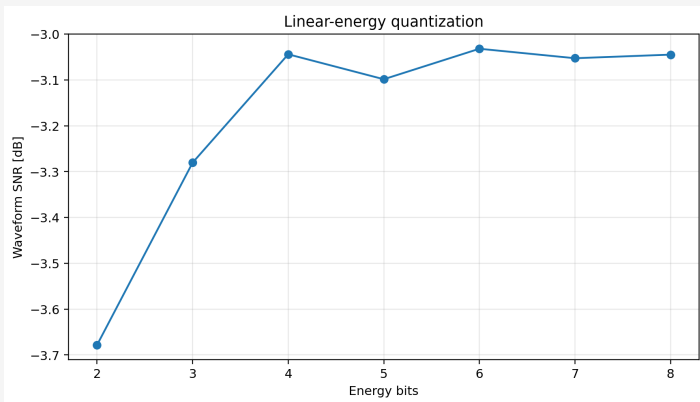


Observed range

$$m \approx 2.29 \times 10^6, \quad x_{\max} \approx 3.14 \times 10^7.$$

The linear-domain energy distribution is strongly right-skewed and not centered at zero; therefore, the offset parameter of the quantizer is important.

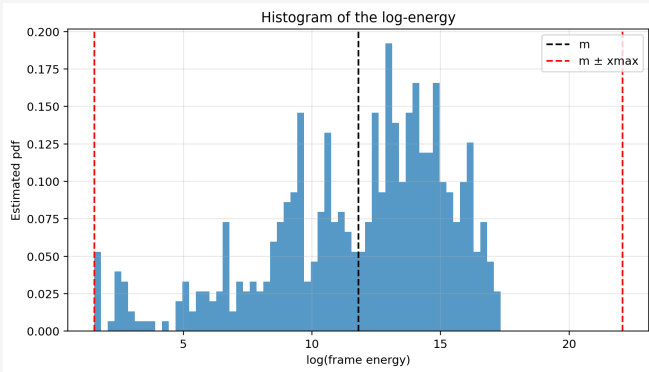
Section 2.1 – Q2: Linear-energy quantization



Final answer

A practical threshold is about **6 bits**. Below that point, the frame-to-frame loudness evolution remains too coarse.

Section 2.1 – Q3: Histogram of the log-energy

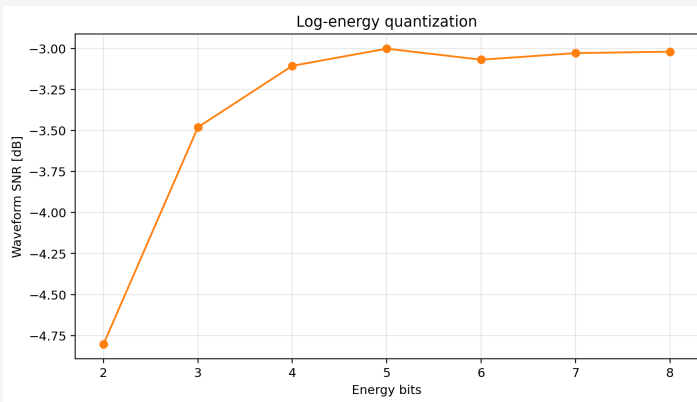


Observed range

$$m_{\log} \approx 11.80, \quad x_{\max, \log} \approx 10.27.$$

The log-energy histogram is much more compact than the linear-energy histogram, which indicates a more efficient use of quantization levels.

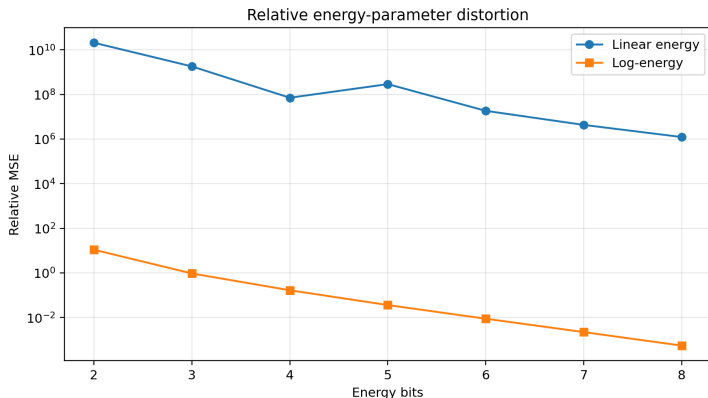
Section 2.1 – Q4: Log-energy quantization



Final answer

A practical threshold is about **5 bits**. The log-domain representation reaches the same perceptual quality with fewer bits than the linear-energy representation.

Section 2.1 – Q5: Linear versus log-energy quantization



Interpretation

The relative MSE is orders of magnitude smaller in the log domain. Therefore, the final conclusion is that **log-energy quantization is preferable to linear-energy quantization** for this task.

Section 2.2 – Q1: Pitch and voiced/unvoiced coding

Efficient encoding scheme

- Pitch-lag range in the analysis: **22 to 133 samples**
- Exact worst-case coding would require **7 bits** on voiced frames.
- Efficient structure:
 - **1 bit/frame** for the voiced/unvoiced decision
 - transmit pitch *only* on voiced frames
- In the final optimized allocation, the pitch quantizer uses **6 bits on voiced frames only**.

Since only about 37% of frames are voiced, this conditional pitch transmission significantly reduces the average bitrate.

Section 2.3 – Q1: LP-parameter quantization by two-stage VQ

The LP coefficients are converted to line spectral frequencies (LSFs), and then quantized in two stages:

$$i_1 = \arg \min_k \|\ell - \mathbf{c}_k^{(1)}\|^2, \quad \mathbf{r} = \ell - \mathbf{c}_{i_1}^{(1)}, \quad i_2 = \arg \min_k \|\mathbf{r} - \mathbf{c}_k^{(2)}\|^2.$$

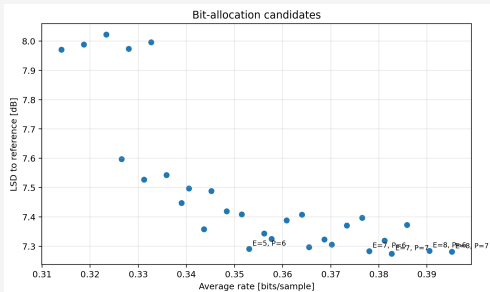
The reconstruction is

$$\hat{\ell} = \mathbf{c}_{i_1}^{(1)} + \mathbf{c}_{i_2}^{(2)}.$$

Final answer

- Each codebook has 1024 entries.
- Stage 1 uses **10 bits**; stage 2 uses **10 bits**.
- Total LP coding rate: **20 bits/frame**.
- Measured LPC coefficient MSE: **0.0128**.

Section 2.4 – Q1/Q2: Bit-allocation search logic



The design was **not** selected by maximizing waveform SNR.

Instead, the main proxy was the **log-spectral distance (LSD)** to the unquantized reference synthesis.

Among the near-best LSD candidates, the chosen operating point was the **lowest-rate configuration**.

Section 2.4 – Q1/Q2: Chosen bit allocation

Final allocation

- Energy: **5 bits** in the log domain
- Pitch: **6 bits** on voiced frames only
- Voiced/unvoiced: **1 bit/frame**
- LP parameters: **20 bits/frame**

This choice preserves the spectral envelope more reliably than a purely waveform-SNR-based selection, while keeping the average bitrate as low as possible among the strongest candidates.

Section 2.4 – Q3/Q4: Final rate and interpretation

Final numeric results

- Waveform SNR relative to the unquantized reference synthesis: **-2.47 dB**
- LSD to the unquantized reference synthesis: **7.29 dB**
- Average bitrate: **0.3530 bits/sample**
- Average bitrate: **2824.1 bits/s**

Interpretation

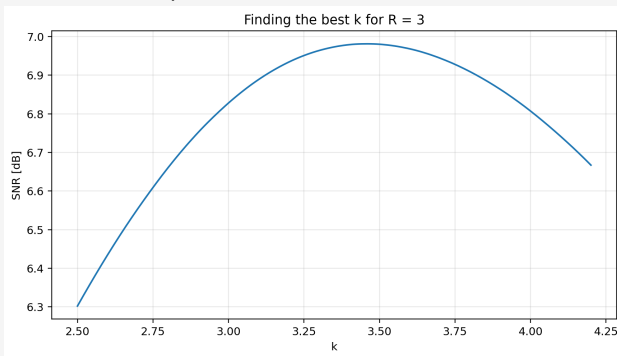
Waveform SNR is not a particularly meaningful quality metric for a vocoder, because the goal is to preserve the spectral envelope, excitation type, pitch, and energy rather than the exact waveform samples. That is why the allocation was selected by a spectral-distance criterion instead.

Section 3 – Q1: Optimizing k for waveform quantization at $R = 3$

The waveform quantizer uses

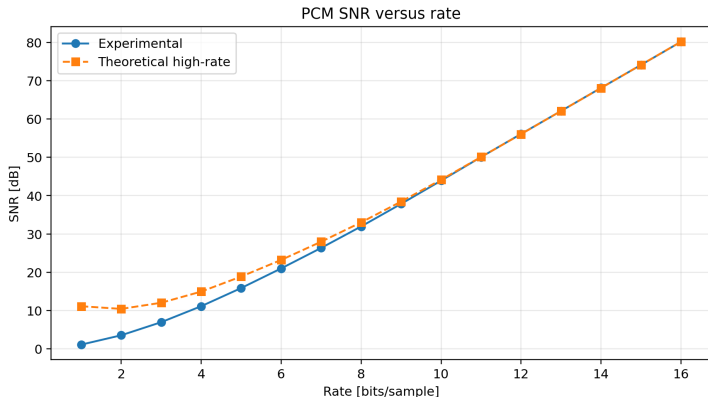
$$x_{\max} = k\sigma_x.$$

The parameter k was swept and the SNR was measured for each value.



Final answer: the SNR-optimal value is $k \approx 3.46$.

Section 3 – Q2/Q3: PCM SNR versus rate and comparison with theory



Interpretation

The experimental SNR becomes almost linear with rate once the quantizer leaves the low-rate regime. The high-rate theoretical expression matches the measured curve very closely from approximately **8 bits/sample** and above.

Section 3 – Q4/Q5/Q6: Perceptual interpretation and low-rate behavior

Questions 4 and 5

- A practical transparency threshold is about **8 bits/sample**.
- At $R = 1$, the error is rough, granular, and strongly signal-dependent.
- As the rate increases, the error becomes weaker and more noise-like.

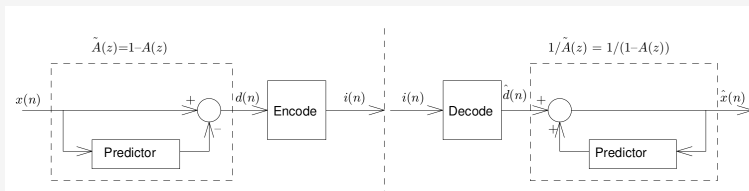
Question 6 (optional)

Measured SNR values:

- $R = 1$: midrise 1.13 dB, midtread 0.99 dB
- $R = 2$: midrise 3.57 dB, midtread 4.56 dB
- $R = 3$: midrise 6.98 dB, midtread 8.65 dB

At low rates, a reconstruction level at the origin becomes beneficial for this speech signal.

Section 4 – Q1: Adaptive open-loop DPCM structure



The predictor operates in open loop, the residual is quantized, and the decoder reconstructs the signal through the synthesis predictor.

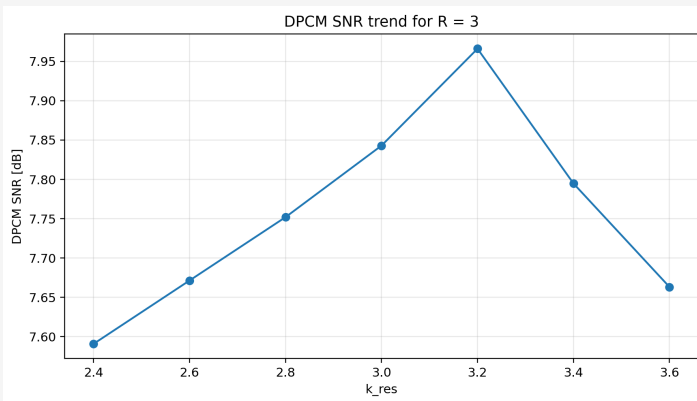
Section 4 – Q1: Chosen DPCM settings

Chosen design

- Frame length: **240 samples**
- Update length: **240 samples**
- Window: **Hann**
- LPC order: **10**
- Residual quantizer: **3 bits/sample**
- Residual scale parameter: **5 bits/frame** in the log domain
- LP parameter coding: **20 bits/frame**
- Encoder predictor uses the **quantized LP coefficients**

The design uses matched LP models in the encoder and decoder so that the prediction and synthesis stages remain consistent after quantization.

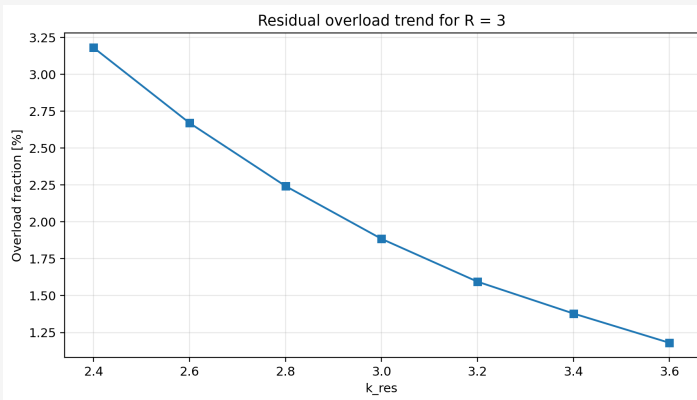
Section 4 – Q1: Evidence for the choice of k_{res} (SNR trend)



Interpretation

The useful operating region is the plateau around $k_{res} = 3.0$ to 3.2 . The final design keeps $k_{res} = 3.0$, because it lies in that plateau while avoiding a larger quantization step than necessary.

Section 4 – Q1: Evidence for the choice of k_{res} (overload trend)



Interpretation

Increasing k_{res} reduces overload distortion, but it also increases the step size. The final choice $k_{res} = 3.0$ is therefore a tradeoff between overload probability and granular-noise level.

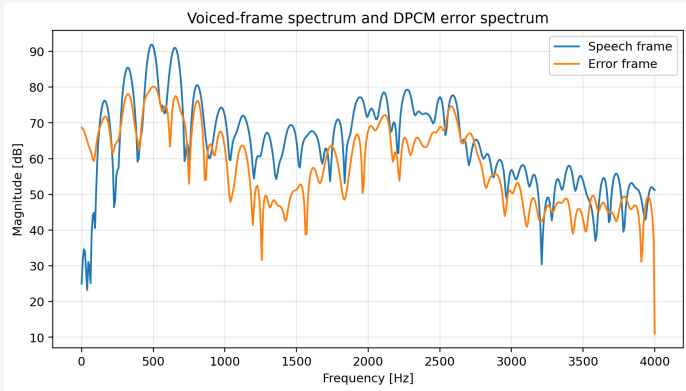
Section 4 – Q2: Reconstructed speech and error characterization

Final answer

- The reconstructed speech is clearly more natural than direct 3-bit PCM.
- The error remains audible, but it is not purely granular; it is shaped by the prediction and synthesis filters.
- Consequently, the error behaves like a colored residual rather than white quantization noise.

These conclusions are consistent with the obtained spectrum on the next slide and with the structure of the DPCM system.

Section 4 – Q3: Error spectrum of a voiced frame



Section 4 – Q3: Theory and interpretation of the error spectrum

Theory and interpretation

If the quantization noise in the residual domain is approximately white, then after the synthesis filter the output-error PSD is shaped by

$$S_{ee}(e^{j\omega}) \propto \frac{1}{|A(e^{j\omega})|^2}.$$

Therefore, the error spectrum follows the spectral envelope of the speech rather than remaining flat.

Section 4 – Q4: DPCM versus PCM at the same rate

Final numeric comparison

- DPCM rate: **3.104167 bits/sample**
- DPCM SNR: **7.84 dB**
- Same-rate PCM by interpolation: **7.42 dB**
- Nearest integer reference (3-bit PCM): **7.00 dB**

The DPCM design outperforms both the interpolated same-rate PCM reference and the nearest integer-rate PCM operating point.

Section 4 – Q5: Total DPCM rate

The total DPCM bitrate is

$$R_{tot} = R_{res} + \frac{R_A + R_\sigma}{N_f} = 3 + \frac{20 + 5}{240}.$$

Hence,

$$R_{tot} = 3.104167 \text{ bits/sample} \quad \Rightarrow \quad R_{tot} = 24833.3 \text{ bits/s.}$$

Final answer

The decomposition is:

- **3 bits/sample** for the residual waveform quantizer
- **20 bits/frame** for the LP parameters
- **5 bits/frame** for the residual scale parameter

Section 4 – Q6: Quantized versus unquantized LP coefficients in the encoder

Measured result

- Encoder with **unquantized** LP coefficients: **2.92 dB**
- Encoder with **quantized** LP coefficients: **7.84 dB**

Conclusion

The encoder predictor should use the **same quantized LP model** as the decoder synthesis filter. This keeps the system matched and avoids a large model-mismatch penalty.

Thank you for your attention.

Questions are welcome.